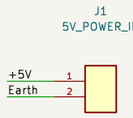
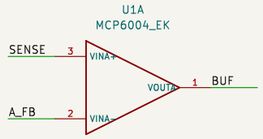
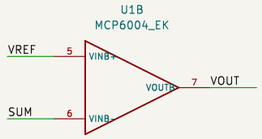
RAS304 | RESISTIVE SENSOR AMPLIFIER | EK | TEAM 103

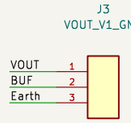






















U1A: unity−gain buffer via R3. U1B: gain = −R5/R4 = −10 about VREF = 2.5V.

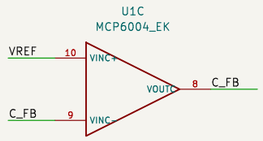
Ideal linear response: VOUT = 27.5 − 10*V1; output clips near the 0V / 5V supply rails.

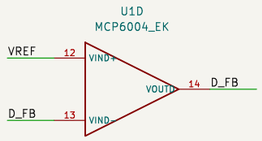
Sensor connects externally across J2 pins 1 and 2. R1 = 5.1k; R5 = 100k (user lab values).











Unused U1C/U1D configured as followers at VREF. C1 bypasses VDD/VSS near U1.





All resistors: through−hole axial, 7.62mm lead spacing. Capacitors: 2.54mm lead spacing.

Supply is externally regulated +5V. Do not connect the earlier 9V supply directly to this PCB.

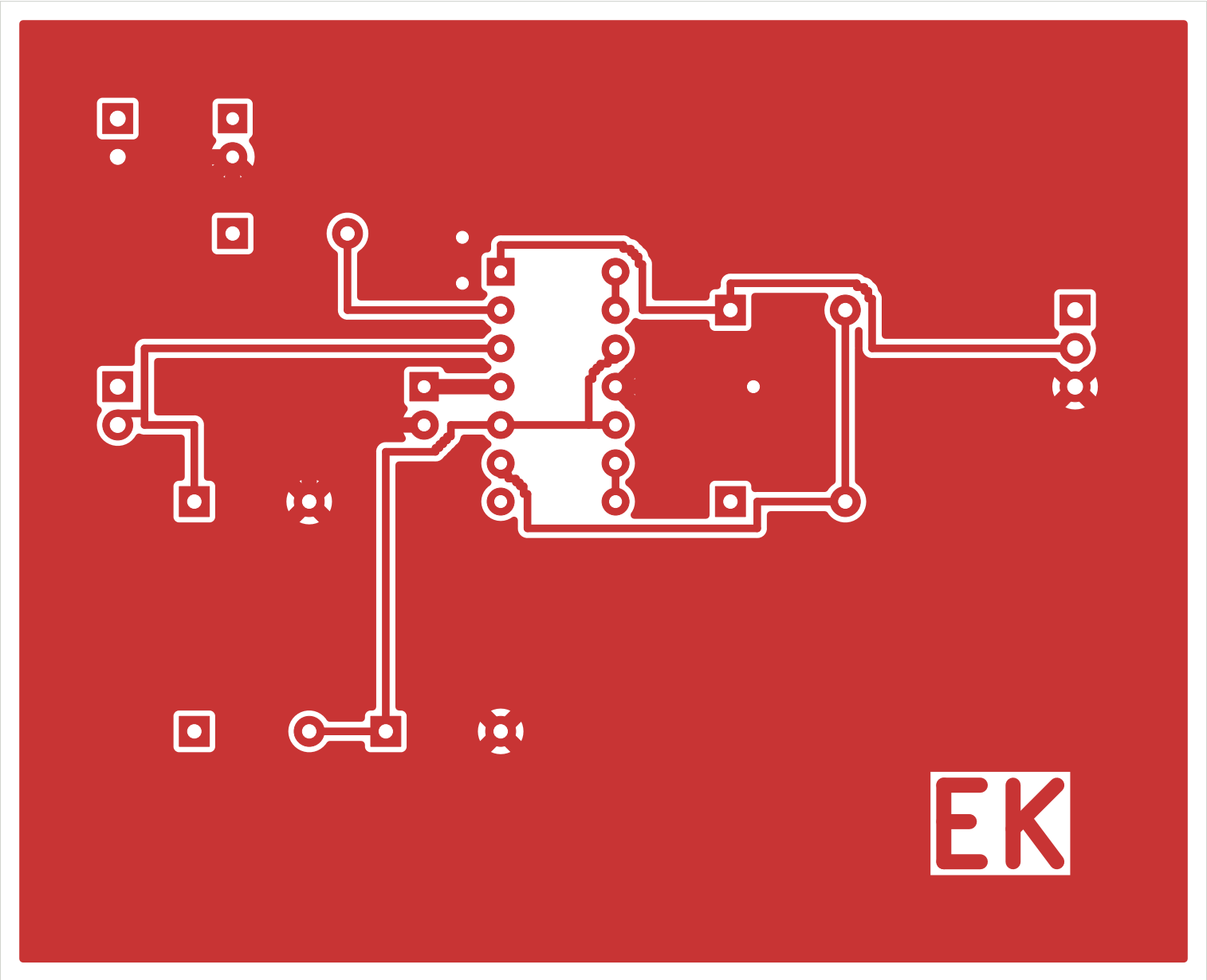
RAS304 − EK

Sheet: /
File: EK_Sensor103.kicad_sch

Title: Resistive Sensor Amplifier − Team 103

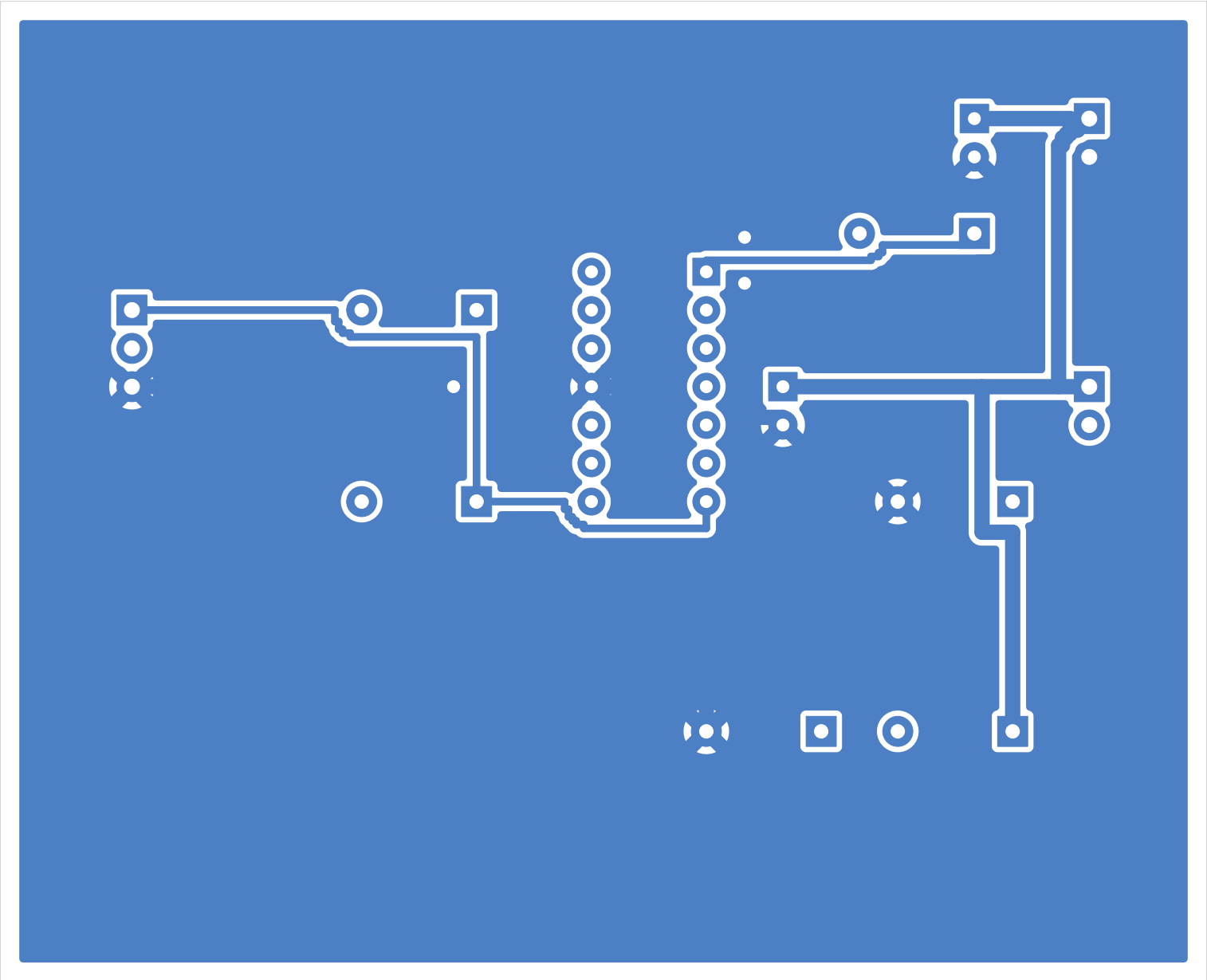
Size: A3	Date: 2026−10−20	**Rev: 1.0**
KiCad E.D.A. kicad−cli 7.0.11+dfsg−1build4		Id: 1/1

EK | Team 103 | Top copper - viewed from above



Native KiCad plot. Board: 80 x 65 mm. Enlarged view; use Assembly_1to1.pdf for physical fit checks.

EK | Team 103 | Bottom copper - viewed from below (mirrored)



Native KiCad plot. Board: 80 x 65 mm. Enlarged view; use Assembly_1to1.pdf for physical fit checks.